

**THE IMPACT OF IMMERSIVE TECHNOLOGIES ON STEM
EDUCATION IN SRI LANKA'S SECONDARY AND TERTIARY
CURRICULUM**

25-26J-418

Project Proposal Report

Hettiarachchi G V – IT22363466

BSc (Hons) in Information Technology Specializing in Interactive Media

Department of Interactive Media

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

We hereby declare that this research proposal is our own independent work carried out as part of our undergraduate research project. It does not embody, without acknowledgement, any work submitted for a degree, diploma, or other academic award at any university or higher institution of learning. To the best of our knowledge and belief, it does not contain any matter previously published or written by some other individual, except where proper acknowledgement has been made in clear terms within the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Date:	Signature of the Supervisor:
Date:	Signature of the Co-Supervisor:

ABSTRACT

How are Sri Lankan students actually going to understand the path of a projectile, the flow of current in a circuit, or the bending of light at the boundary if their classrooms offer only static textbook diagrams and little lab equipment? These abstract and dynamic phenomena are prone to be abstruse, therefore leaving Grade 7 learners depended upon memorization. Augmented Reality (AR) promises a new path—making textbook images real with interactive 3D models. International research (Rahmat et al., 2023; Arymbekov & Turekhanova, 2025) reports significant learning gains with AR, but the technologies have had scant influence on the Sri Lankan classroom, where inexpensive Android devices dominate. This research proposes a Unity AR app aligned with local syllabus, utilizing marker tracking to animate textbook diagrams and facilitate real-time variable manipulation. In testing the effect in comparison with traditional methods, this research asks: could AR be the means to render abstract physics tangible, engaging, and memorable learning?

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LIST OF ABBREVIATIONS

AR	Augmented Reality
ARKit	Apple AR Development Kit (For IOS)
ARCore	Google AR Development Kit (For Android)
AI	Artificial Intelligence
SDLC	Software Development Life Cycle
WBS	Work Breakdown Structure

Table i : List of Abbreviations

INTRODUCTION

There are particular difficulties in teaching physics to junior secondary students. At this point, students are moving from their common ideas about the physical world to abstract, scientific representations. Students must mentally model dynamic, invisible, or multi-dimensional processes in order to understand concepts like projectile motion, energy transfer, light refraction, and basic electric circuits. However, traditional teaching methods in Sri Lanka still mainly rely on rote memorization and static textbook illustrations, leaving little opportunity for experimentation, visualization, or inquiry. Globally, this pedagogical gap has long been recognized as a barrier to physics students' deep conceptual learning (Arymbekov & Turekhanova, 2025).

Virtual reality (VR) and augmented reality (AR) technologies have become revolutionary tools in education over the last 20 years. They are becoming more and more common in science classrooms, according to a thorough bibliometric analysis of AR/VR in physics education (Prahani et al., 2022), which showed a steady upward trend in publications between 2002 and 2021. The most often used keywords were augmented reality, physics education, students, and education computing, indicating a global understanding of AR's potential to improve understanding and engagement. However, another study also found that South Asia, an educational region with limited resources, was underrepresented in research efforts, which were mostly focused in technologically advanced regions.

Empirical studies also bear witness to AR's potential in fostering achievement and motivation in physics. For instance, Rahmat et al. (2023) carried out a six-week quasi-experiment on 64 junior-high students in Indonesia and reported that AR-supported instruction in astronomy resulted in significantly higher normalized learning gains than textbook-based instruction, with students characterizing AR as a "new learning environment" rendering abstract content concrete. Likewise, Arymbekov & Turekhanova (2025) found that AR-based visualization software supported intellectual activity by enabling students to adjust variables and see the results immediately, which was not achievable with static visuals. These studies highlight AR's capability in rendering abstract

phenomena concrete through manipulable 3D simulations, which strongly resonates with the needs of Grade 7 physics students.

However, previous efforts also demonstrate a number of limitations. First and foremost, most programs assume the presence of high-end hardware and uninterrupted internet connectivity, a situation that is unrealistic within the Sri Lankan learning environment, where most students use low-end Android smartphones and face intermittent connectivity. In addition, even as studies in augmented reality (AR) have emphasized student motivation, they neglect the preparedness of instructors and curriculum alignment. According to Arymbekov and Turekhanova (2025), the effective adoption of AR requires sustained professional development and pedagogical support systems, which are rarely incorporated into available prototypes. Moreover, despite a growing corpus of AR studies in physics from different parts of the world, Prahani et al. (2022) expressly reflect the absence of localized case studies from developing countries, thus reflecting a geographical lacuna in the literature that the present project seeks to fill.

To overcome the aforementioned weaknesses, the current study aims to develop a Unity-based AR application for the Sri Lankan Grade 7 physics syllabus. In contrast to prior research that has addressed astronomy or general science content, this project will integrate AR modules into textbook diagrams themselves (ray diagrams, trajectory paths, circuit diagrams), thus guaranteeing syllabus relevance and classroom practicality. The system will be designed to work optimally offline and on low-end devices, bypassing infrastructural constraints. Additionally, this project will not only assess student learning gains via a quasi-experimental study design but also offer teacher-facing guides and training resources to support adoption.

In short, existing work shows AR's potential for physics education but raises unanswered questions regarding scalability in low-resource settings, alignment with national curricula, and sustainability via teacher buy-in. By placing this project in the context of Sri Lankan Grade 7, this research seeks to close that gap—providing both a technological platform and empirical findings to move the state of the art in physics education research forward.

1.1 Background & Literature Survey

Augmented Reality (AR) has increasingly evolved from the novelty technology that it was to a pedagogical device for making abstract, dynamic physics phenomena more concrete for school-age learners. In classrooms for middle-grade students, in which students are transitioning from phenomenological description to formal representations, AR's capacity for mapping manipulable, three-dimensional representations onto familiar contexts promises greater immediacy of correspondence between students' mental models and conventional physics representations. Contemporary summaries for middle-school educators emphasize that effective science instruction increasingly depends on the insights of cognitive science and educational technology, and that research-based, peer-reviewed materials (journals, professional associations, lesson banks) are critical to translating those insights into teaching practice—context that justifies the new proposal's emphasis on research-congruent AR design as opposed to "technology for technology's sake."

World trends: AR/VR in physics education, 2002–2021

A 20-year bibliometric survey (Scopus-indexed) demonstrates steady production in AR and VR research in education, of which physics learning is a consistent sub-topic. The survey identifies "augmented reality," "students," "education computing," and "physics learning" as leading co-occurring phrases in AR-physics publication, indicating a sustained focus on classroom integration and student performance. The review further notes that, while interest is growing globally (and in the U.S., China, and Germany, where total AR/VR production is currently dominant), much of the AR-in-physics research is still churning through computer science and engineering venues—an imbalance that could continue to under-represent K-12 and teacher-education perspectives. Importantly, the authors cite pragmatic issues of AR/VR stability and optimization as recurring themes throughout most of the most-cited studies—a limitation of engineering with direct implications for school deployment. Implication for this project. The Grade 7 Unity-based AR component needs to prioritize didactic congruity (curricular objectives, assessment) while engineering for toughness on mid-range mobile devices, which means paying

attention to the "stability/optimization" problem mentioned in the most-cited work in the field.

Empirical evidence with junior-secondary learners

Classroom studies from junior-secondary classrooms reveal that mobile AR improves mastery of abstract physics concepts over textbook learning alone.

In a convergent-parallel, mixed-methods design with 64 seventh-graders over six weeks (West Java, Indonesia), students who learned using a mobile AR app on a solar system unit learned more (assessed via pre/post testing and normalized-gain scores) than controls with paper materials; interviews described AR as a "new learning environment" that grounds 3D concepts and makes them understandable. The studies describe pragmatic constraints typical of lower-secondary settings—first AR deployment, in-classroom use of smartphones, short unit duration—that align well with the present project's constraints and population. Implication for the project at hand. For Grade 7 physics (such as forces and motion, energy transfer, and optics), an AR experience emphasizing manipulable 3D objects aligned with assessment items should yield measurable short-term gains, since the intervention is nested within a definite pre/post evaluation structure and augmented with brief onboarding of students to the AR interface.

AR versus VR for school physics: design and teacher-readiness

A more recent comparative review argues that, in learning environments, AR is more readily improved for engagement and retention than VR, again highlighting two system designs: (i) preparing teachers for AR education, and (ii) student participation in co-creating AR content to tailor it for relevance and usability. Synthesis literature of the study emphasizes the strength of AR in visualizing and solving complex concepts of physics without eliminating context from real life entirely—an advantage for transfer of concepts in K-12 classrooms.

These results, along with the documented teacher-accessibility problems for immersive technologies, justify the present proposal's AR-first strategy for Grade 7 and teacher-facing guides and classroom-management assistance. Implication for this project. Beyond the student experience, the Unity build should provide lightweight teacher materials (usage flow, misconception prompts, troubleshooting) and a professional-learning micro-module, directly filling the teacher-readiness gap found in the literature. Middle school pedagogy and resource ecosystem For middle-school science, resource letters and professional organizations (AAPT, NSTA) mandate research-tested practices and journals that embed cognitive principles in classroom practices. These sources point to inquiry-driven tasks, multiple representations, and technology-enhanced visualization as the key levers of conceptual change—guiding design decisions like scaffolding from formal to real-world representations and embedding formative checks. Embedding an AR tool in this setup increases adoption likelihood and instructional fidelity.

Synthesis for the aimed Grade 7 AR component (Unity, mobile)

Through trend analysis, classroom testing, and practitioner resources, three design imperatives emerge:

1. Pedagogical alignment and measurability. Focused intervention addressing widely acknowledged Grade 7 areas of pain (abstract, dynamic phenomena) and match the AR interactions with pre/post tests and misconception-related prompts, mirroring approaches in previous junior-secondary research.
2. Stability and classroom readiness. Design for stability on typical school hardware and short deployments; design for low onboarding and offline-capable assets, in awareness of the optimization problems typical of AR/VR research.
3. Teacher competency and capacity. Provide just-in-time support and micro-PD to overcome the teacher-readiness barrier and maximize the likelihood of long-term, high-fidelity adoption.

Collectively, the literature testifies to the potential and likely effectiveness of a Unity-based AR module for Grade 7 physics, with it being evident that the way the experience is built, facilitated, and supported will be more important than the novelty of the medium itself.

1.2 RESEARCH GAP

The use of Augmented Reality (AR) to enhance the learning of physics has been widely investigated across the globe, and considerable evidence demonstrates that it has a positive impact on students' conceptual grasp and motivation. Nevertheless, a close review of available studies reveals a few gaps that are left unaddressed, and this proposal particularly aims to address those shortcomings in the Sri Lankan context.

i. Gap in Curricular Alignment

- What previous research did

Rahmat et al. (2023) instructed Indonesian Grade 7 students with AR and estimated large learning gains ($\langle g \rangle = 0.71$ compared to 0.30 for conventional instruction).

Arymbekov & Turekhanova (2025) confirmed AR stimulates greater intellectual activity throughout Grades 7–11, with a bias toward exploratory learning.

- What was missing:

None of the studies overlaid AR simulations onto the illustration of a national curriculum textbook, precluding classroom use directly.

There is no official syllabus in Sri Lanka for Grade 7 physics apart from studying textbooks, and therefore it needs to be aligned with a set syllabus.

- How this proposal fills the gap:

The suggested AR app will accept Sri Lankan textbook Grade 7 diagrams straight as markers, thus offering for simple integration to the curriculum as stipulated.

ii. Technological Gap (Low-resource Contexts)

- What research so far accomplished:

A majority of the studies employed AR on relatively well-endowed devices or in urban schools with greater infrastructure capacity.

Arymbekov & Turekhanova (2025) and Rahmat et al. (2023) showed AR's practicality in schools with device availability.

- What was lacking:

No research was done on entry-level Android devices that are pervasive among Sri Lankan students, especially rural and semi-urban schools.

There are not many applications that have used offline-first features, although internet connectivity in schools continues to be poor.

- How this proposal addresses the gap:

The 2–3GB RAM optimized Unity-built AR app will have preloaded assets to ensure it is utilized offline and thus is scalable and viable in the Sri Lankan context.

iii. Empirical Validation Gap (Sri Lankan Context)

- What prior research achieved:

Rahmat et al. (2023) empirically confirmed AR's effectiveness through normalized gain analysis.

Arymbekov & Turekhanova (2025) established engagement and cognitive learning improvement.

Prahani et al. (2022) meta-analyzed 139 AR/VR studies, which determined worldwide effectiveness.

- What was lacking:

None of these studies employed AR in Sri Lankan classrooms, where education is highly exam-driven, practical experiments are not the norm, and teachers are not familiar with AR-based teaching.

- How this proposal fills the gap:

A quasi-experiment study will be conducted with Sri Lankan Grade 7 students, thus generating the first empirical dataset to determine the effectiveness of AR in this particular setting.

iv. Pedagogical Integration Gap (Teacher Readiness)

- What earlier research achieved:

Arymbekov & Turekhanova (2025) recognized AR's potential being maximized by the training of teachers.

Krusberg (2009) stressed inquiry-based teaching but without the integration of AR.

- What was absent:

Precisely very little research on how teachers adopt and integrate AR in lesson plans.

Insufficient supporting resources for instructors (e.g., training modules, ready-to-use guides).

- How this proposal fills the gap:

Along with the development of the AR app, the project will deliver teacher-centered quick-start guides and sample lesson plans in order to facilitate adoption instead of novelty implementation.

v. Deficits in Longitudinal and Regional Studies

- What prior research achieved:

Prahani et al. (2022) traced the path of AR/VR research worldwide (2002–2021), showing strong growth in Asia and Europe.

- What was absent:

Sri Lanka is not represented in this international dataset. Most of the reviewed interventions were short-term pilots rather than systematically embedded in curriculum practice. • How this proposal fills the gap: Through localizing AR into the Sri Lankan Grade 7 curriculum, this research adds the international data in an underrepresented nation and delivers a replicable, scalable model for expansion in the future.

SYSTEM GAP ANALYSIS										
Technical Implementation Comparison - AR Physics Education Systems										
System / Feature	Unity Engine Support	Vuforia/ARCore Integration	Offline Asset Storage	Low-RAM Optimization	Real-time Physics Engine	Marker Recognition	Cross-Platform Deployment	User Progress Tracking	Teacher Dashboard	Curriculum API
Mobile AR Physics (Sabari et al.)	~	✓	✗	✗	✓	✓	✗	✗	✗	✗
PhET Simulations (Web-based)	✗	✗	~	✓	✓	✗	✓	~	✗	✗
Merge EDU (Cube-based AR)	✓	✓	✓	✗	~	✓	~	✓	~	✗
CoSpaces Edu (Block-based AR)	✓	✓	~	✗	~	~	✓	✓	✓	✗
Augmented Physics Constructor (ARToolkit & TuriVision)	~	✓	✗	✗	✓	✓	✗	~	✗	✗
Google Expeditions (Screen-based)	✓	✓	~	✓	✗	✓	✓	✗	~	✗
Proposed Unity AR System (Kollurachchi GV)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Figure 1 : System Gap Analysis

RESEARCH GAP ANALYSIS						
Augmented Reality in Physics Education - Sri Lankan Context						
Application Features	Curriculum Alignment with National Standards	Low-Resource Device Compatibility (2-3GB RAM)	Offline Functionality	Teacher Training Materials	Interactive 3D Physics Simulations	Marker-Based AR Integration
Research A (Rahmat et al., 2023)	X	X	X	X	✓	✓
Research B (Krusberg, 2009)	X	X	X	✓	X	X
Research C (Arymbekov & Turekhanova, 2025)	X	X	X	X	✓	✓
Research D (Prahani et al., 2022)	X	X	X	X	X	X
Research E (Existing AR Platforms)	X	X	X	X	✓	X
Proposed System (Hettharatchchi G.V.)	✓	✓	✓	✓	✓	✓

Figure 2 : Research Gap Analysis

1.3 RESEARCH PROBLEM

Lower secondary physics education in Sri Lanka, by way of Grade 7 science education, is characterized by several recurring problems hindering students' concept understanding and memorization for the long term. The education system relies heavily on textbooks, where abstract concepts such as projectile motion, simple electric circuits, refraction of light, and wave propagation are addressed with static two-dimensional diagrams. While these diagrams are to be utilized as additional illustrative tools, they are inherently limited in describing dynamic processes, unseen forces, and multi-dimensional interactions. For this reason, students are generally not capable of developing accurate mental models of physical phenomena, leading to misconceptions, memorization by rote, and ineffective application of knowledge in problem-solving contexts.

Coupled with this is the reality that in the majority of schools, especially in rural and semi-urban areas, there are not enough proper laboratory facilities. There are limitations in experimental equipment, and hence students are restricted from hands-on learning, which is a prerequisite for learning physics fundamentals. The teachers too are constrained by conventional teacher-centered pedagogical methodologies that are more inclined to test-

oriented learning than conceptual learning. Thus, there is a pedagogical lacuna where theory is more emphasized but practice and visualization needn't be adequately developed.

Globally, Augmented Reality has been recognized as a novel instrument in transcending such barriers by enabling interactive three-dimensional visualization of intangible concepts. Studies such as Rahmat et al. (2023) and Arymbekov & Turekhanova (2025) validate AR as a means to enhance learning outcomes and motivate intellectual engagement in learning physics. These studies were conducted in educational environments with different curricular structures and comparatively enhanced technological structure. As Prahani et al. (2022) noted, AR/VR research has grown globally, but in Sri Lanka, no empirical research exists. In addition, existing AR applications usually demand top-of-the-line hardware and stable internet connectivity, which are not the features of the technology infrastructure of Sri Lankan classrooms where most students use low-spec Android devices and frequently face connectivity issues.

Thus, the research problem is two-fold:

1. Pedagogical Issue: How do Grade 7 Sri Lankan students overcome the limitations of static textbook teaching and under-resourced laboratory facilities for learning physics?
2. Technological Issue: How can AR be best designed and developed to function pragmatically in the limits of the poor-resource school context of Sri Lanka while being directly aligned with the national curriculum?

This research will address these concerns through the development of a Unity-powered AR app that is tailored to the Sri Lankan Grade 7 physics syllabus to be operated on low-spec Android devices with offline capabilities. Apart from development, the study will investigate the impact of instruction using AR-based resources compared to traditional instructional methods empirically using a quasi-experimental approach, thereby addressing a significant research gap in localized context. In doing so, it hopes to transform physics education in Sri Lanka from being primarily an exercise in memorization to one of visualization, interaction, and conceptual understanding.

2 OBJECTIVES

2.1 Main Goal

Based on the Grade 7 physics curriculum in Sri Lanka, the goal is to develop, implement, and validate a Unity-based Augmented Reality (AR) application that will enhance students' conceptual grasp of abstract physical concepts through interactive three-dimensional visualization, remain affordable for low-end Android devices, and be sustainable through teacher integration.

2.2 Specific Goals (SMART Framework)

1. Curriculum Alignment (specific, attainable):

To look into Sri Lanka's Grade 7 physics curriculum and find difficult subjects that can't be represented with conventional techniques, like projectile motion, fundamental electrical circuits, light refraction, and wave propagation, and incorporate them into syllabus-aligned augmented reality simulations that are triggered straight from textbook illustrations.

2. AR Application Development (Specific, Measurable, Achievable):

To make and release a Unity-based AR smartphone app that lets students see and change physics principles (like changing launch angles, resistance values, or refractive indices) with instant results. The app should work well on phones with 2–3GB of RAM and be able to work offline without needing the internet.

3. Teacher Support & Pedagogical Integration (Realistic, Achievable):

To create teacher-centered materials-ranging from quick-start manuals, exemplary lesson plans, and training micro-modules of AR-that facilitate the integration of AR into pedagogy, overcoming potential barriers to teacher uptake and ensuring sustainability beyond pilot status.

4. Empirical Evaluation (Measurable, Time-bound):

To execute a quasi-experimental study within 8–10 weeks of Sri Lankan Grade 7 classes, contrasting an experimental group (instruction based on AR) with a control group (traditional teaching), assessing students' learning gains via pre- and post-tests (normalized gain $\langle g \rangle$) and evaluating students' motivation and engagement through surveys and interviews.

5. Testing for performance and usability (measurable and doable):

To assess the application's technical performance (responsiveness, tracking accuracy, device compatibility) and usability (user-friendliness, interface, student/teacher satisfaction) via systematic usability testing and feedback sessions.

6. Contribution of Knowledge (Time-bound, Realistic):

To produce initial empirical findings regarding the efficacy of augmented reality in physics education in Sri Lanka and to enhance global literature by addressing the deficiency of South Asian case studies as discussed in Prahani et al. (2022). The findings will be documented and disseminated via research outputs and stakeholder reports throughout the project duration.

3 METHODOLOGY

3.1 General System Description

The study will be conducted through a Design–Develop–Evaluate (DDE) framework, structured around the development of a strategic partnership between Unity-enabled AR technology and the Sri Lankan Grade 7 physics curriculum. The system will utilize marker-based AR technology, where students scan textbook diagrams (printed pictures) with their mobile phone, and the system overlays interactive 3D models of physics concepts.

The strategy merges three simultaneous strands:

- I. System Development (Technical Implementation) – building the AR app in Unity with Vuforia/AR Foundation.
- II. Pedagogical Integration (Educational Design) – mapping to the Sri Lankan Grade 7 physics curriculum and production of teacher support materials.
- III. Evaluation (Research Design) – empirical field testing of the app in real classrooms through quasi-experimental procedures.

3.2 Technologies to be Used

The main game engine for the Physics AR component will be Unity 3D, which will make use of AR Foundation 5.0 and Vuforia for AR rendering and marker-based tracking. Simulation logic, parameter manipulation, and user interface interactions will all be implemented using the C# programming language. Unity's integrated Physics Engine will perform physics computations to guarantee accurate representation of motion, light refraction, and electrical circuits. The application will run smoothly in classroom settings and be optimized for low-end Android devices, guaranteeing offline functionality.



3.3 System Architecture

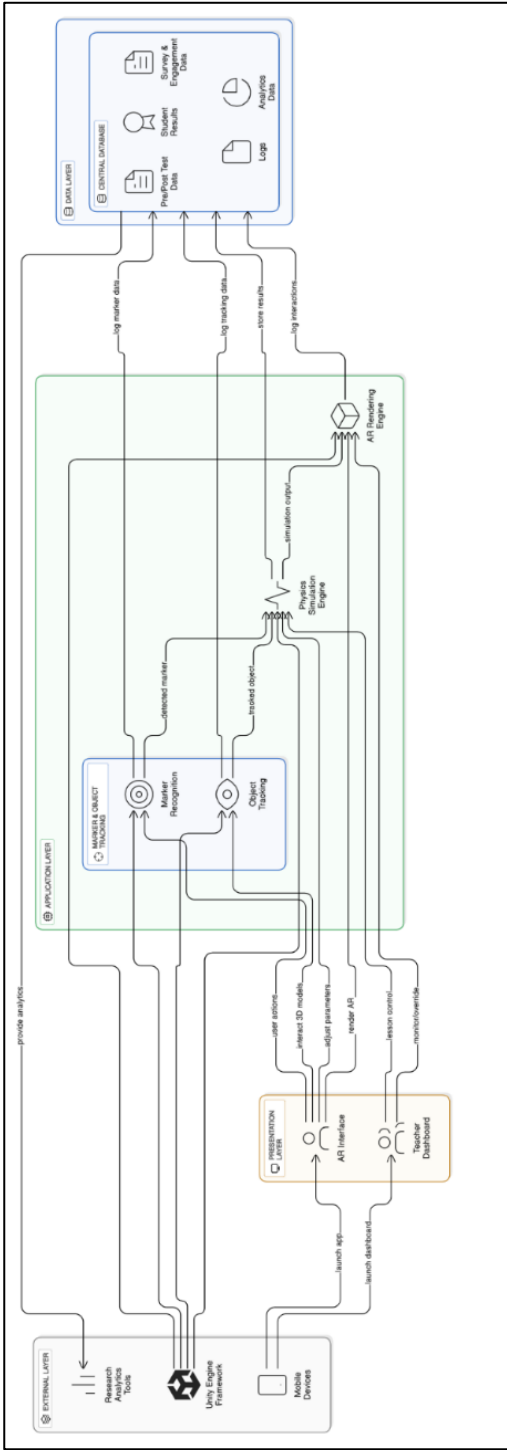


Figure 3 : System Architecture

3.4 Research Design

The quasi-experimental study design will involve two groups:

- Experimental Group: Students study seventh-grade physics concepts using the AR app.
- Control Group: Using traditional textbook-style instruction, students are taught the same material.

Participants: About 60 seventh-graders from two urban and semi-urban Sri Lankan schools took part in order to achieve diversity.

Approach:

- i. Pre-test: Both groups take a conceptual understanding test as a baseline assessment.
- ii. Intervention (6–8 weeks): The experimental group receives AR-augmented lessons in both lab and classroom environments. The control group receives textbooks and instruction in the form of lectures.
- iii. Post-test: The same test is administered to both groups in order to measure normalized gain.
- iv. Qualitative feedback: informal conversations with teachers and students to assess adoption barriers, usability, and engagement.

Methods of Data Collection:

- Quantitative: Normalized improvement in learning analysis and conceptual knowledge pre- and post-tests.
- Qualitative: Focus groups with students, teacher interviews, and classroom observations.
- Technical: System logs (interaction time, frequency of AR usage).

3.5 Development & Testing Phases

- I. Requirement Analysis (Weeks 1–2): Identify physics topics, define AR learning goals, analyze device constraints.
- II. System Design (Weeks 3–4): Create wireframes, build system diagram, storyboard user interactions.
- III. Prototype Development (Weeks 5–8): Unity development with simple AR modules (circuit behavior, projectile motion).
- IV. Pilot Testing (Week 9): Small group testing (10–15 students) for performance and usability.
- V. Full Deployment (Weeks 10–15): Classroom implementation with experimental and control groups.
- VI. Data Analysis (Weeks 16–18): Compare test scores, review logs, conduct interviews.
- VII. Reporting (Weeks 19–20): Write research findings, prepare documentation.

3.6 Expected Outcomes and Practical Application

When compared to students who use traditional textbook-only methods, students who use the AR app will demonstrate higher normalized learning gains. Reduction of misunderstandings regarding important physics concepts in Grade 7, like circuit behavior, light refraction, and projectile motion. Instructors will give constructive criticism regarding the usability of the classroom, citing better instruction, increased student participation, and simple concept explanation. With support for offline use and low-end Android devices, the AR app will show useful scalability for Sri Lankan schools with limited infrastructure. Increased readiness of educators to incorporate augmented reality (AR) technology into their lesson plans as a regular teaching tool.

3.7 System Diagram

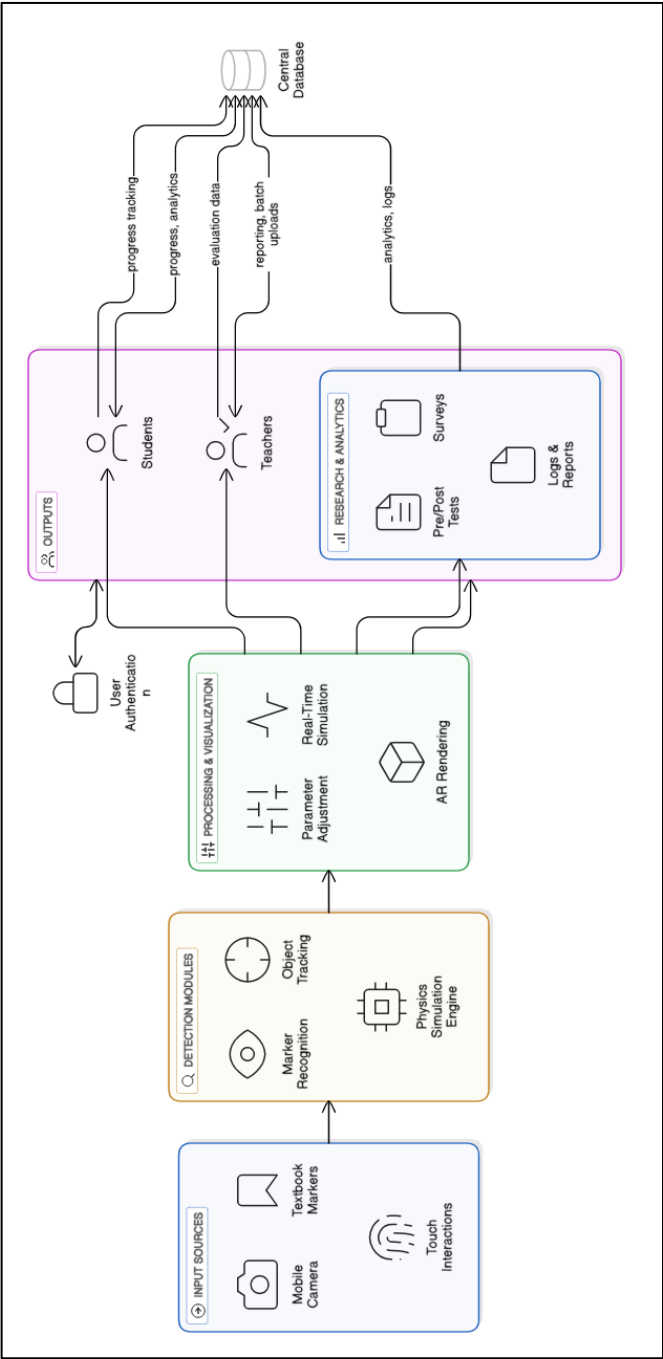


Figure 4: Overall System Diagram

4 PROJECT REQUIREMENTS

4.1 Functional Requirements

These are the primary features and functionality that the AR application should support in order to accomplish its goals.

i. Marker-based Recognition:

The system should be able to recognize Grade 7 physics textbook diagrams (e.g., circuit symbols, ray diagrams, trajectory graphs) through the device camera.

After recognition, the app should appropriately place the corresponding 3D simulation on top of it.

ii. Interactive 3D Simulations:

The software must render principles of physics as manipulable 3D objects (e.g., a projectile whose path changes when launch angle is manipulated, a circuit with variable resistance/voltage).

Manipulation of variables in real time must be facilitated for students (sliders, toggle switches, gestures).

iii. Curriculum Alignment:

The simulations must align one to one with Sri Lankan Grade 7 physics syllabus topics, so that they are applicable for class teaching.

iv. Offline Functionality:

All assets (3D models, textures, simulations) must be preloaded, thus enabling the application to operate offline.

v. User Profiles (Optional future extension):

Save individual user activity logs (time spent, simulations viewed).

Data export for teachers (CSV/Excel) for student engagement analysis.

vi. *Teacher Support Tools:*

Provide quick-access guides within the application (short "how to use" tips).

Teachers' ability to select or turn off which AR simulations are available for a lesson.

4.2 Non-Functional Requirements

These define quality aspects and system performance criteria.

i. *Performance:*

The AR system must be smooth on low-end Android devices (min 2GB RAM, quad-core processor).

Marker recognition response time ≤ 2 seconds in ordinary lighting.

ii. *Scalability:*

The system must allow for future growth (e.g., including Grade 8/9 subjects without recompiling the core engine).

iii. *Usability:*

Interface must be intuitive, with minimal text and simple controls appropriate for 12–13-year-old students.

Must adhere to UI/UX best practice (large buttons, small icons, clear instructions).

iv. *Reliability & Robustness:*

App must be reliable in a variety of classroom environments (lighting conditions change, markers partially occluded).

Should include fallback instructions ("Marker not detected, adjust camera") when recognition cannot be performed.

v. *Portability:*

System needs to work with Android devices supporting Android 8.0 (Oreo) and above.

Adaptability to future iOS versions should be considered (via Unity cross-platform support).

vi. Maintainability:

Code must respect modular design principles (e.g., possess AR recognition, simulation engine, and UI modules separated)).

(Documentation and comments must allow for easy modifications.

vii. Security & Privacy:

(The application must not gather or transmit student private information without consent.

(Offline design ensures respect for low-data environments and privacy rules).

5 Work Breakdown Chart

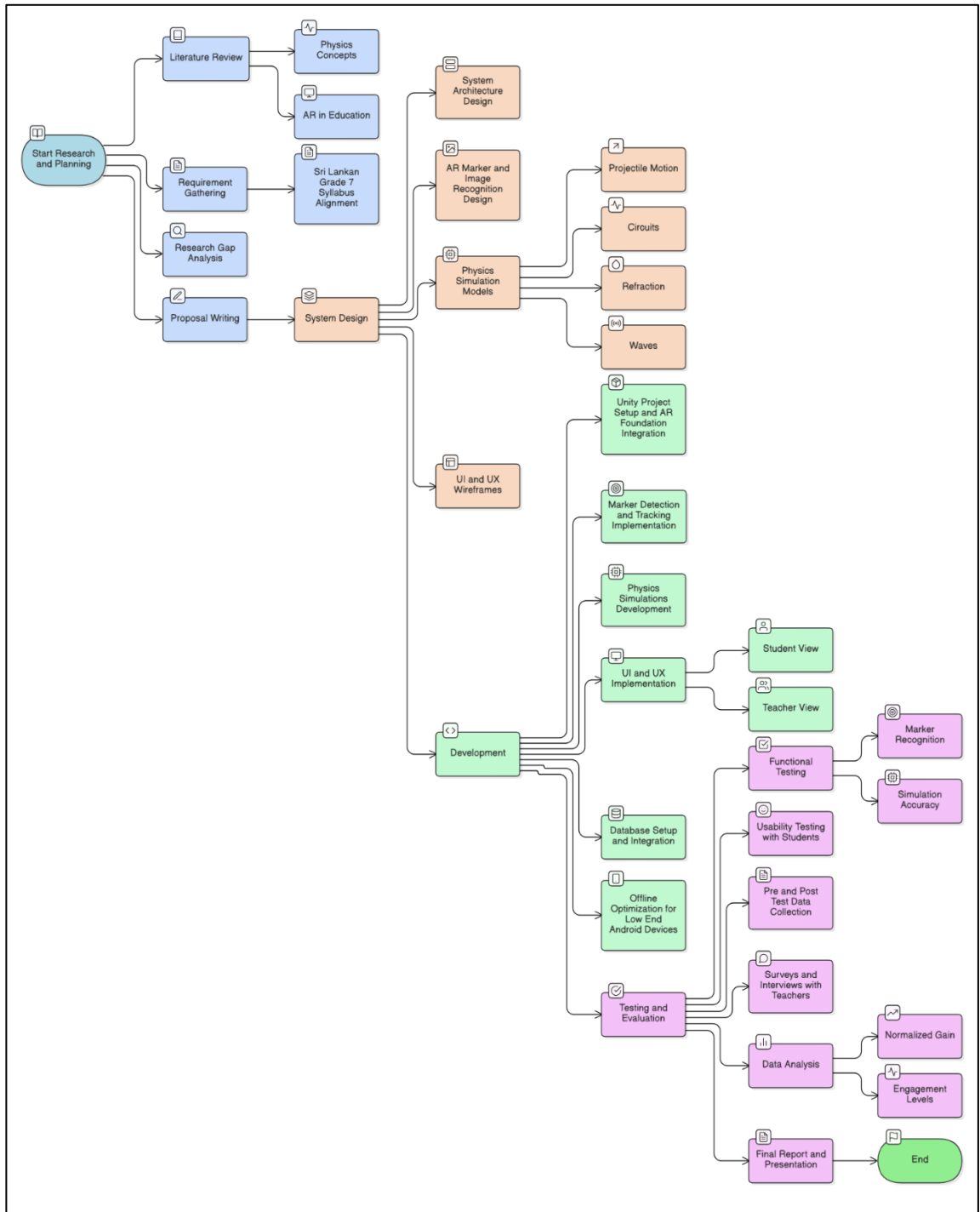


Figure 5 : Work Breakdown Chart

6 PROJECT TIMELINE

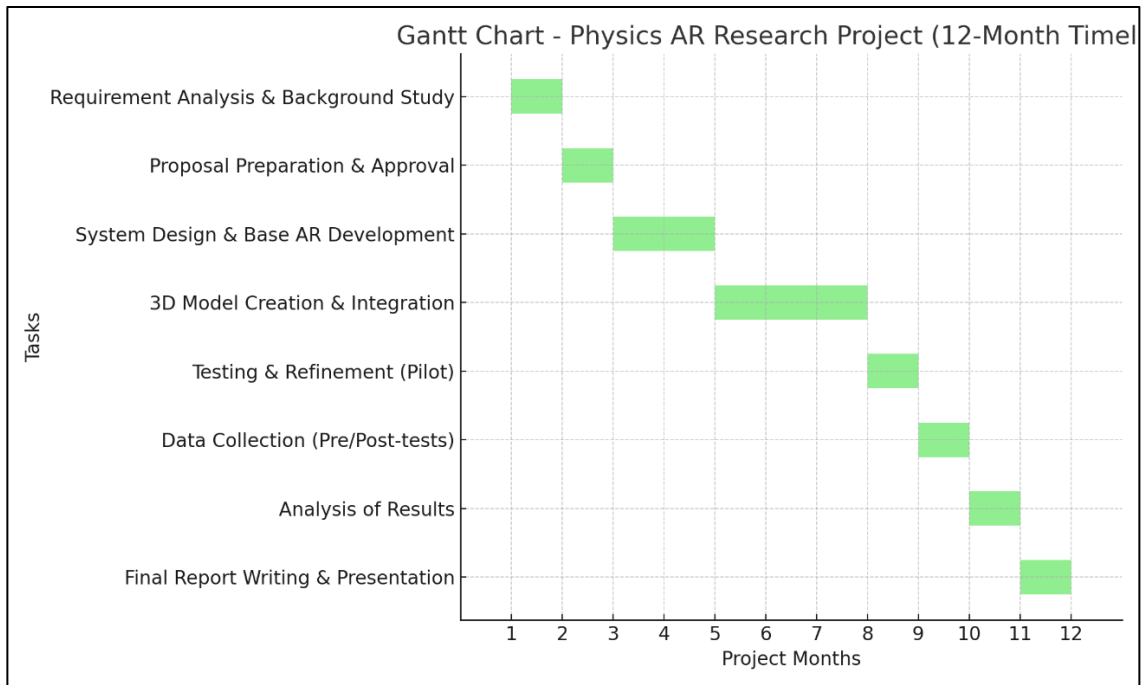


Figure 6 : Gantt Chart

7 BUDGET AND BUDGET JUSTIFICATION

Investment Category	Cost Breakdown (LKR)	Pricing & Recovery Model (LKR)
Labor Costs	Development: LKR 2,500,000 Research: LKR 800,000 Testing: LKR 500,000	Freemium Model: Free: Basic Grade 7 content Premium: LKR 900/month full access
Material Costs	3D Assets: LKR 600,000 Software Licenses: LKR 300,000 Hardware/Testing: LKR 400,000	Institutional Licensing: School License: LKR 60,000/year District License: LKR 300,000/year
Marketing & Distribution	Digital Marketing: LKR 800,000 Partnership Development: LKR 500,000	Government Contracts: Ministry of Education: LKR 15,000,000/year Provincial Councils: LKR 4,500,000/year each

Table ii : Budget Allocation

8 REFERENCES APPENDICES

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